Amdad Ahmed Awsaf, EdS
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EDUCATION

PhD, STEM Education August 2022 – July 2027 (expected)
CGPA 4.00/4.00 (to date)
Florida International University, Miami, FL
Dissertation Advisor: Dr. Remy Dou

EdS in Teaching and Learning, STEM Education August 2022 – April 2025
CGPA 4.00/4.00
Florida International University, Miami, FL

Master of Education (M.Ed.) January 2020 – July 2022
Science, Mathematics, and Technology Education
CGPA 4.00/4.00
University of Dhaka, Bangladesh

Bachelor of Education (B.Ed.) January 2016 – December 2019
Science, Mathematics, and Technology Education
CGPA 3.80/4.00
University of Dhaka, Dhaka, Bangladesh

RESEARCH

Graduate Assistant, Florida International University (FIU) May 2026 – Present
The DREAM Center
Advisor: Dr. Elizabeth Cramer

- Supporting the federally funded OSEP technical assistance center focused on strengthening special education personnel preparation under IDEA
- Assisting in development and organization of institutional reports and materials supporting program improvement and dissemination
- Contributing to documentation and synthesis of center activities for educator preparation, training, and technical assistance initiatives

Graduate Research Assistant, Florida International University (FIU) August 2025 – Present
Advisor: Dr. Zahra Hazari, PhD

Research Projects: STEP UP: Supporting Teachers to Encourage the Pursuit of Undergraduate Physics

- Analyze quantitative and qualitative data related to interventions aimed at increasing physics engagement and participation pathways
- Contribute to preparation of manuscripts and research reports for dissemination of project findings
- Co-author grant proposals supporting ongoing and future research initiatives in physics education and STEM participation

Graduate Research Assistant, Florida International University (FIU) August 2023 – August 2025
Advisor: Dr. Remy Dou, PhD

Research Projects:

1. CAREER: Talking Science: Early STEM Identity Formation Through Everyday Science Talk" (NSF Award AISL #1846167)

2. OYE Tumble: Originating Youth Engagement with Culturally-Situated Science Podcasts for Kids (AISL #2415575)

- Examined youths' STEM identity development, emphasizing community cultural capital and everyday science discourse in informal learning contexts
- Designed and refined research instruments (e.g., surveys and semi-structured interviews) grounded in theory-driven research questions and stakeholder input
- Conducted stakeholder and participant interviews and supported field-based data collection in informal STEM education settings
- Transcribed, managed, and coded qualitative data in alignment with human subjects research protocols and ethical standards
- Analyzed qualitative and quantitative data to investigate patterns in STEM engagement and identity development using RStudio
- Mentored and trained two undergraduate research assistants in data collection, transcription, analysis, and research procedures
- Contributed to dissemination activities, including preparation of research reports and support for conference presentations and project deliverables for NSF-funded work

Graduate Research Assistant, Center for Astrophysics | Harvard & Smithsonian
Advisor: Dr. Philip M. Sadler, Dr. Remy Dou, Dr. Sue Sunbury Dec. 2022 – July 2024

- Researched the impact of community cultural capital on STEM engagement
- Designed surveys, conducted interviews, and analyzed qualitative and quantitative data
- Transcribed interviews and handled confidential research data
- Assessed informal education programs to inform policy decisions

Project: "*Racially and Ethnically Minoritized Youths' Out-of-School-Time Experiences and Their Effects on STEM Attitudes, Identity, and Career Interest*" (NSF Award AISL #2215050)

Skills: RStudio · Interviewing · Survey Design · Statistical Data Analysis · Handle Confidential Information · Database Management System (DBMS) · Analytical Skills · Qualitative Research

Master of Education Thesis July 2022
- Nature and impact of informal Professional Learning Community's technological support during COVID-19 in enhancing secondary science teachers' efficacy

Early Career Researcher (ECR), University of Dhaka March 2022 – July 2022
- As an ECR for the *3Mpower-Mobile Learning for Empowerment of Marginalised Mathematics Educators* research project, I visited the field, conducted teacher interviews, and collected and analyzed data

Research Assistant (RA), University of Dhaka February 2022 – June 2022
- As an RA at the *Effectiveness of alternative assessment approaches in higher education during pandemic: Perception of stakeholders,* I have conducted faculty and student interviews
- Conducted surveys and analyzed qualitative and quantitative data

TEACHING

Instructor of Record, Florida International University (FIU Spring, 2025
SMT4664 Educational Assessment and Project-Based Instruction in Mathematics and Science
- Led instruction and assessment for Florida Educator Accomplished Practices (FEAP)-aligned teacher certification course focused on STEM assessment and project-based instruction
- Co-designed curriculum and performance-based assessments with faculty mentor
- Evaluated 11 pre-service teachers

- Supervised 5 pre-service teachers in school-based practica and conducted site visits to 3 high schools for instructional evaluation

Graduate Teaching Assistant, Florida International University (FIU) Fall, 2022
EDF 7404 Qualitative Research Data Analysis

- Assisted in course design and instructional delivery for a graduate-level qualitative research data analysis course
- Facilitated and supported instructional activities and in-class learning under faculty supervision

Intern Teacher, Udayan Higher Secondary High School, Dhaka January 2019 – June 2019

- Instructed mathematics and science at the secondary level (middle and high school equivalent)
- Developed and implemented subject-based lesson and unit plans aligned with curriculum standards
- Assessed student learning through grading and evaluation of examinations and project-based work

PUBLICATIONS

1. **Awsaf, A. A**., Dou, R., Cian, H., & Secules, S. (2026). A Systematic Literature Review of Survey Research and Measurement of Engineering Identity. *European Journal of Engineering Education* (in preparation)

2. **Awsaf, A. A.,** Hazari, Z., Sabouri, P., Mateu, C. (2026) Examining how teachers' disciplinary science identity and formal/informal professional development experiences relate to their students' disciplinary major intentions: A physics exemplar. *Journal of STEM Teacher Education* (in preparation)

3. **Awsaf, A.A**., Dou, R., & Gerhard, S., & Sadler, P. (2026) Pre-College Recognition Experiences and STEM Identity Development Among College Youths: An Extended Structural Equation Model. *International Journal of STEM Education* (in preparation)

4. Nti-Asante, E., **Awsaf, A. A.** (2025). *Constructing* Hybrid Mathematical Discourse: A Third-Space Approach to Ghanaian Dialects and English. *African Journal of Research in Mathematics, Science and Technology Education* (*Revised and Resubmitted*)

5. **Awsaf, A.A**., Siddique, M., N., A., Rahman, S., M., H. (2026). Examining the Impact of Informal PLCs' Technological Support on Science Teachers' Efficacy: An Exploratory Sequential Mixed Method Study. *Teacher Development* (Under review)

CONFERENCE PRESENTATIONS (Peer-reviewed)

1. **Awsaf, A.A.**, Dou, R., Sunbury, S., Gerhard, S., & Sadler, P. (2026, April). *Carrying Contextual Learning Experiences Forward: A Structural Equation Model for Youths' STEM Identity Development* [Presentation]. 2026 AERA Annual Meeting, Los Angeles, California.

2. **Awsaf, A.A.**, Dou, R., Sunbury, S., Gerhard, S., & Sadler, P. (2026, April). A Structural Equation Model for Youths' STEM Identity Development: Impact of Contextual STEM Learning Experiences [Presentation]. 2026 NARST Annual International Conference, Seattle, Washington.

3. **Awsaf, A.A.**, & Irwin, C. (2026, April). Testing Cross-National Measurement Invariance of TIMSS 2023 EAB Framework and Implications in Environmental Education Research

[Presentation]. 2026 NARST Annual International Conference, Seattle, Washington.

4. **Awsaf, A.A.**, Dou, R., Gerhard, S., & Sadler, P. (*2025*, *April*). *Examining the impact of perceived experiences on STEM identity and career aspirations among underrepresented youth* [Presentation]. 2025 AERA Annual Meeting, Denver, CO.

5. **Awsaf, A.A.**, Giasanti, N., Sunbury, S., & Dou, R. (*2025*, *March*). *Concurrent phenomenological analysis of STEM career aspirations in underrepresented youth: Role of experiences and identity* [Presentation]. 2025 NARST Annual International Conference, National Harbor, MD.

6. **Awsaf, A.A.**, Dou, R., Gerhard, S., & Sadler, P. (*2025*, *March*). *Exploring associations between event-based (mis)recognition by STEM authorities with STEM identity and career aspirations* [Presentation]. 2025 NARST Annual International Conference, National Harbor, MD

7. **Awsaf, A.A.**, Cian, H., & Dou, R. (2024, March). *A Systematic Literature Review of Survey Research on Engineering Identity*[Presentation]. 2024 NARST Annual International Conference, Denver, CO.

8. **Awsaf, A. A.**, Siddique, M. N. A., & Rahman, S. M. H. (October 2023). *Impact of Informal PLC's Technological Support During COVID-19 in Enhancing Secondary Science Teachers' Efficacy*. Southeastern Association for Science Teacher Education Annual Conference, Miami, Florida.

OTHER PRESENTATIONS:

1. Kim, J., **Awsaf, A.A.**, & Irwin, C. (2026, April). Testing Cross-National Measurement Invariance of TIMSS 2023 EAB Framework and Implications in Environmental Education Research [Presentation]. Graduate Student Appreciation Week (GSAW) Scholarly Forum 2026, FIU, Miami, FL.

2. **Awsaf, A.A.**, Dou, R., Sunbury, S., Gerhard, S., & Sadler, P. (2026, April). A Structural Equation Model for Youths' STEM Identity Development: Impact of Contextual STEM Learning Experiences [Presentation]. Graduate Student Appreciation Week (GSAW) Scholarly Forum 2026, FIU, Miami, FL.

3. **Awsaf, A.A.**, Dou, R., Gerhard, S., & Sadler, P. (2025, April). STEM Identity and Career Aspirations of College Youths: The Role of Event-Based (Mis)Recognition by STEM Authorities. *Poster presented at SEHD conference*, Miami, FL.

INVITED TALKS:

1. **Awsaf, A.A.** (2025, January). NISE Net Online Workshop: STEM Identity & Career Interest for Students Traditionally Underrepresented in STEM. National Informal STEM Education Network, Virtual, https://vimeo.com/1048979874?fl=pl&fe=cm.

HONORS AND AWARDS

- 2026 – Jhumki Basu Fellowship Award
- 2026 – Second Place Award, GSAW Scholarly Forum 2026, Florida International University
- 2025 – NARST Graduate Student Committee Travel Scholarship
- 2022 – Bangladesh Government Scholarship, Granted by Ministry of Education, Bangladesh

PROFESSIONAL MEMBERSHIPS

- National Association for Research in Science Teaching (NARST)
- American Educational Research Association (AERA)

EXTRA-CURRICULAR AND COMMUNITY SERVICES

Examination Invigilator, British Council Dhaka December 2020 – May 2022

- Conducted the A-level and O-level exams of Pearson, Edexcel, and Cambridge
- Invigilated in the IELTS exam

Intern, BRAC Education Program, Dhaka October 2021 – February 2022

- Worked with the content team to research best practices and training in content development
- Reviewed books and other materials for new content and curriculum development
- Assisted the training team to conduct the assessment, evaluation, and follow-up with trainees

Intern, Agami Education Foundation March 2018 – December 2018
Worked as a Science and Mathematics content localizer at the *Khan Academy Bangla project*

SKILLS

Computer: SPSS, RStudio, NVivo, MAXQDA, Atlas.ti
Language: Bangla, English, Spanish (elementary proficiency)